



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

30-05-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS & TNAS]

ANSWER KEY

PAPER-1

### PART-1 : PHYSICS

|                 |    |         |         |       |       |      |      |
|-----------------|----|---------|---------|-------|-------|------|------|
| SECTION-I (i)   | Q. | 1       | 2       | 3     | 4     |      |      |
|                 | A. | D       | B       | C     | D     |      |      |
| SECTION-I (ii)  | Q. | 5       | 6       | 7     | 8     | 9    | 10   |
|                 | A. | A,B,C,D | A,B,C,D | A,B   | A,C   | B,C  | B,D  |
| SECTION-II (i)  | Q. | 1       | 2       | 3     | 4     | 5    | 6    |
|                 | A. | 8.00    | 38.00   | 26.67 | 12.50 | 1.33 | 4.00 |
| SECTION-II (ii) | Q. | 7       | 8       | 9     |       |      |      |
|                 | A. | 7       | 3       | 6     |       |      |      |

### PART-2 : CHEMISTRY

|                 |    |         |       |       |        |      |         |
|-----------------|----|---------|-------|-------|--------|------|---------|
| SECTION-I (i)   | Q. | 1       | 2     | 3     | 4      |      |         |
|                 | A. | C       | D     | C     | B      |      |         |
| SECTION-I (ii)  | Q. | 5       | 6     | 7     | 8      | 9    | 10      |
|                 | A. | A,B,C,D | A,B,D | B,D   | A,B    | A    | A,B,C,D |
| SECTION-II (i)  | Q. | 1       | 2     | 3     | 4      | 5    | 6       |
|                 | A. | 3.00    | 3.00  | 91.00 | 140.00 | 3.00 | 5.00    |
| SECTION-II (ii) | Q. | 7       | 8     | 9     |        |      |         |
|                 | A. | 5       | 4     | 6     |        |      |         |

### PART-3 : MATHEMATICS

|                 |    |       |       |      |      |      |       |
|-----------------|----|-------|-------|------|------|------|-------|
| SECTION-I (i)   | Q. | 1     | 2     | 3    | 4    |      |       |
|                 | A. | D     | D     | C    | C    |      |       |
| SECTION-I (ii)  | Q. | 5     | 6     | 7    | 8    | 9    | 10    |
|                 | A. | A,C   | A,B   | D    | B,C  | A,C  | A,C,D |
| SECTION-II (i)  | Q. | 1     | 2     | 3    | 4    | 5    | 6     |
|                 | A. | 10.00 | 15.00 | 8.00 | 2.00 | 0.50 | 96.00 |
| SECTION-II (ii) | Q. | 7     | 8     | 9    |      |      |       |
|                 | A. | 429   | 2     | 4    |      |      |       |

### HINT – SHEET

#### PART-1 : PHYSICS

##### SECTION-I (i)

1. Ans (D)

$$l = \frac{m\ell^2}{12} = 20 \times \frac{3}{4} \times \frac{3}{4} \times \frac{1}{12} = \frac{180}{192}$$

2. Ans (B)

$$P = \frac{V^2}{4R} \Rightarrow R = \frac{V^2}{4P} = \frac{48 \times 48}{4 \times 200} = 2.88\Omega$$

3. Ans (C)

$$K_{eq1} = 2(8k \cos^2 30^\circ) + 2(8k \cos^2 60^\circ)$$

$$= 12k + 4k = 16k$$

$$\text{So, } K_{eq} = \frac{(16.16)}{32} k = 8k$$

4. Ans (D)

$$\frac{10}{100} \times 40 = 5 - \frac{5}{R+1} \times 1 \Rightarrow R = 4\Omega$$

## PART-1 : PHYSICS

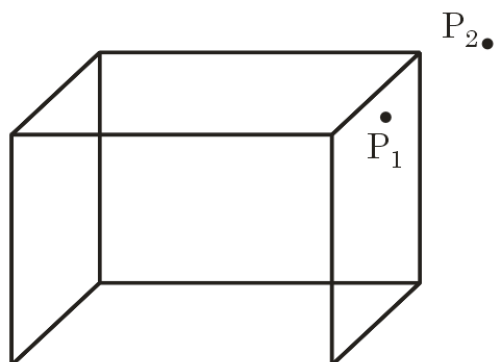
### SECTION-I (ii)

5. Ans ( A,B,C,D )

Using symmetry if charged particle lies at  $P_1$

then  $\phi_{\text{OCDE}} = \frac{q}{24\epsilon_0}$  and

$$\phi_{\text{OABC}} = \frac{\left(\frac{q}{\epsilon_0} - \frac{q}{8\epsilon_0}\right)}{3} = \frac{7q}{24\epsilon_0}$$

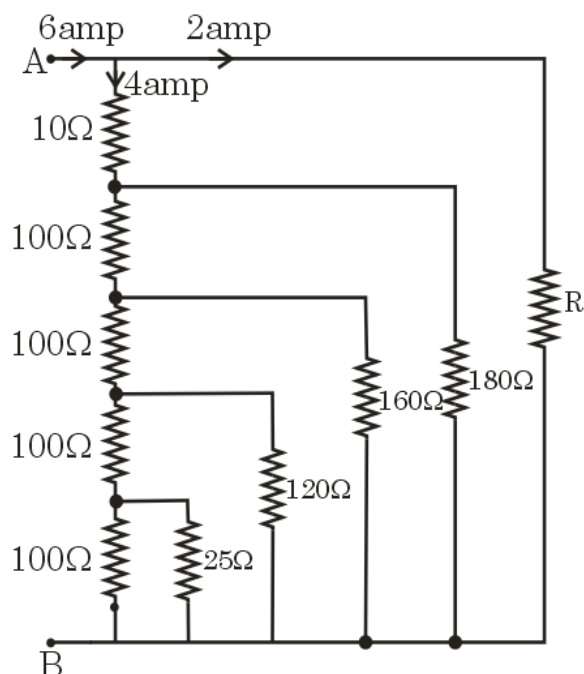


If the charge particle lies at  $P_2$  then

$$\phi_{\text{OCDE}} = \frac{q}{24\epsilon_0}$$

$$\text{and } \phi_{\text{OABC}} = \frac{q}{24\epsilon_0}$$

6. Ans ( A,B,C,D )



Equivalent resistance between P and Q =  $100\ \Omega$ .

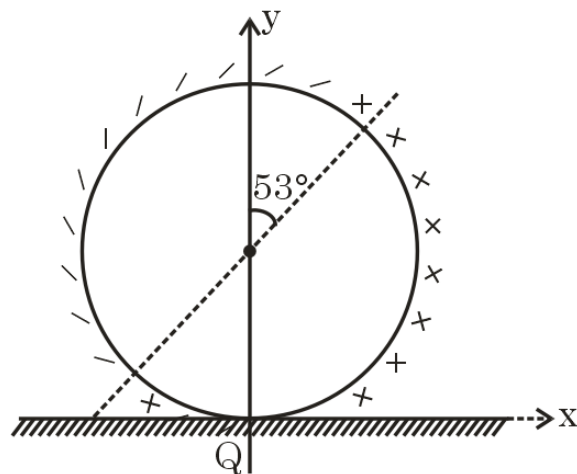
So current towards P =  $\frac{400}{100} = 4$  ampere

Current through R =  $6 - 4 = 2$  ampere

$$\text{So, } R = \frac{400}{2} = 200\Omega$$

7.

**Ans ( A,B )**



$$\tau_Q = PE \sin 37^\circ$$

$$= (\sigma 2\pi R^2) (R) E_0 \frac{3}{5}$$

$$= \frac{6}{5} \sigma \pi R^3 E_0 = \left( \frac{5}{3} m R^2 \alpha \right)$$

$$\alpha = \frac{18}{25} \frac{\sigma \pi R E_0}{m}$$

8.

**Ans ( A,C )**

Angular velocity of earth  $\omega_1 = \frac{\pi}{12}$  rad/hr

Angular velocity of satellite  $\omega_2 = \frac{\pi}{3}$  rad/hr

$$(\omega_2 - \omega_1)t = 2\pi$$

t = 8 hours

→ for option 'B'

$$T^2 \propto r^3$$

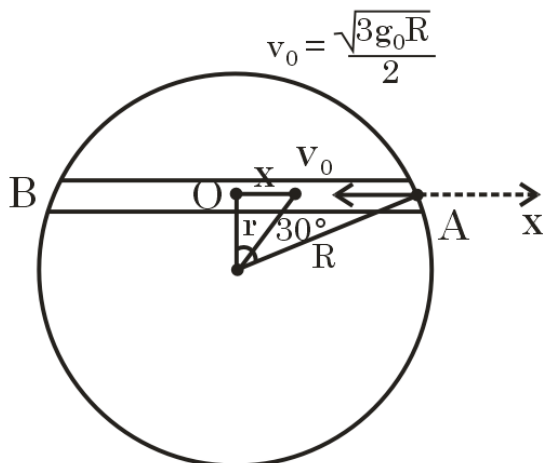
T' = 48 hours

$$\omega' = \frac{\pi}{24} \text{hours}$$

t = 48 hours

9.

Ans (B,C)



$$\frac{d^2x}{dt^2} = -\frac{GM r \cos \theta}{R^3} = -\left(\frac{GM}{R^3}\right)x = -\left(\frac{g_0}{R}\right)x$$

$$\omega = \sqrt{\frac{g_0}{R}}$$

$$\text{Now, } x = A \sin(\omega t + \alpha)$$

$$\text{at } t = 0, x = +R/2$$

$$\frac{R}{2} = A \sin \alpha \dots\dots(i)$$

$$V_x = \omega A \cos(\omega t + \alpha)$$

$$\therefore X = R \sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$$

$$-\frac{R}{2} = R \sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$$

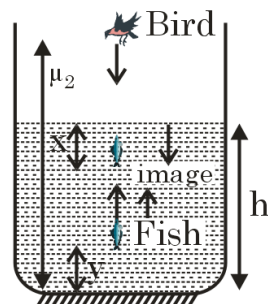
$$\frac{7\pi}{6} = \sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}$$

$$\therefore t = \frac{\pi}{3} \sqrt{\frac{R}{g_0}}$$

$$V_{\max} = \omega A$$

$$V_{\max} = \sqrt{\frac{g_0}{R}}R = \sqrt{g_0 R}$$

10. Ans (B,D)



Distance of image of fish as seen by bird only after refraction is

$$Z = (x - h) + (h - y) \frac{\mu_2}{\mu_1}$$

$$Z = (x - h) + (h - y) \frac{1}{4/3}$$

Differentiate w.r.t. time

$$\frac{dZ}{dt} = (-6 + 2) + (-2 - 4) \frac{3}{4} = -\frac{17}{2} \text{ cm/sec}$$

Similarly we can calculate for 2<sup>nd</sup> case.

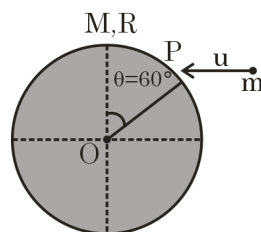
## PART-1 : PHYSICS

### SECTION-II (i)

2.

Ans (38.00)

Using conservation of angular momentum of the system about a horizontal axis passing through centre O.



$$\mu R \cos \theta = \left(\frac{MR^2}{2} + mR^2\right) \omega_0$$

$$\mu R \cos 60^\circ = \left(\frac{M}{2} + m\right) R^2 \omega_0$$

$$0.5 \times 20 \times \frac{1}{2} = 2.5 \times \frac{1}{4} \times \omega_0$$

$$\Rightarrow \omega_0 = 8 \text{ rad/s}$$

Now, using conservation of energy of the system

$$mgR(1 - \cos \theta) = \frac{1}{2} \left(\frac{MR^2}{2} + mR^2\right) (\omega_0^2 - \omega^2)$$

$$mg \frac{R}{2} = \frac{1}{2} \left(\frac{M}{2} + m\right) R^2 (\omega_0^2 - \omega^2)$$

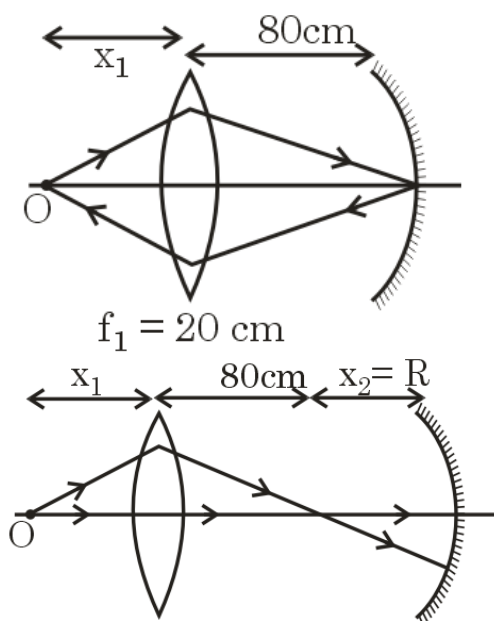
$$0.5 \times \frac{10}{2} = \frac{5}{4} \times \frac{1}{4} (\omega_0^2 - \omega^2)$$

$$\Rightarrow \omega_0^2 - \omega^2 = 8$$

$$\omega^2 = \omega_0^2 - 8 = 64 - 8 = 56$$

$$\omega = \sqrt{56} \text{ rad/s}$$

4. Ans ( 12.50 )



$$\frac{1}{80} - \frac{1}{-x_1} = \frac{1}{20}$$

$$\frac{1}{x_1} = \frac{1}{20} - \frac{1}{80} \Rightarrow x_1 = 26.67 \text{ cm}$$

$$x_2 = R = 12.50 \text{ cm}$$

6. Ans ( 4.00 )

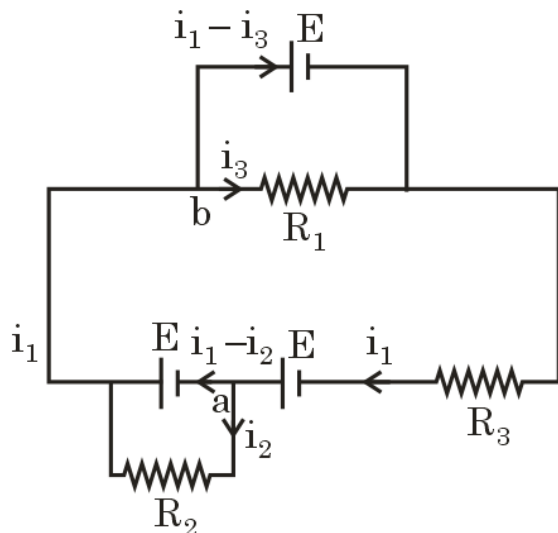
On solving by KVL we get,

$$i_1 = \frac{E}{R}, i_2 = \frac{E}{R}, i_3 = \frac{E}{R}$$

$$\Rightarrow i_1 - i_2 = \frac{2E}{R} \text{ and } i_1 - i_3 = 0$$

So reading of ammeter  $A_1$  is  $E/R = 4/3$

$A_2$  is  $2E/R = 8/3$ ,  $A_3$  is zero and  $\Delta V_{ab} = 4V$



## PART-1 : PHYSICS

### SECTION-II (ii)

7. Ans ( 7 )

$$\text{Speed of projection } (v_0) = v_0 = \sqrt{\frac{2GM}{R}} \dots\dots(1)$$

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r} \dots\dots(2)$$

$$v = \sqrt{\frac{2GM}{r}}$$

$$\int_R^{4R} \sqrt{r} dr = \sqrt{2GM} \int_0^t dt$$

$$t = \frac{7}{3} \sqrt{\frac{2R^3}{GM}}$$

8. Ans ( 3 )

$$q_0 = \int \rho dv = \int_0^R \rho_0 r^n 4\pi r^2 dr$$

$$q = \int \rho dv = \int_0^{R/2} \rho_0 r^n 4\pi r^2 dr$$

$$\frac{kq_0}{R^2} = 16 \cdot \frac{kq}{\left(\frac{R}{2}\right)^2}$$

So  $n = 3$

9. Ans ( 6 )

Velocity of image :

$$8 \times \frac{2}{4/3} \times \frac{1}{2} = 6 \text{ cm/s}$$

## PART-2 : CHEMISTRY

### SECTION-I (i)

3. Ans ( C )

Part of theory

4. Ans ( B )

If FeO is like NaCl (rock salt)

Then no. of  $\text{Fe}^{2+}$  per unit cell will be = 4

crystal  $\text{Fe}_{0.75}\text{O}$  contain some  $\text{Fe}^{2+}$  (x) & some  $\text{Fe}^{3+}$  (0.75-x) so  $2x + (0.75 - x) \times 3 = 2$

$$\Rightarrow x = \frac{1}{4} \Rightarrow \text{number of } \text{Fe}^{3+} \text{ per unit cell} = 1 \times 2$$

## PART-2 : CHEMISTRY

### SECTION-I (ii)

5. **Ans ( A,B,C,D )**

Correct order regarding atomic size :

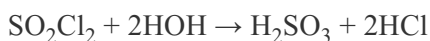
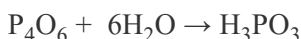
(A)  $Zr \approx Hf$  (due to lanthanide contraction)

(B)  $Fe \approx Co \approx Ni$  ( $Z_{eff} \approx$  Screening effect)

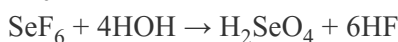
(C)  $Sc < Y < La$  (Shells increasing)

(D)  $Cu < Zn$  ( $Z_{eff} <$  Screening effect)

6. **Ans ( A,B,D )**



$SF_6 \rightarrow$  no hydrolysis



$\Rightarrow$  all are dibasic oxyacid

9. **Ans ( A )**

If  $T_1 = T_2 = T_3$  then molecular mass of gas 'A' > molecular mass of gas 'B' > molecular mass of gas 'C'.

10. **Ans ( A,B,C,D )**

Moles of A = 10 moles

Moles of B = 40 moles

$$\therefore \text{Mole fraction} \Rightarrow x_A = \frac{10}{50}$$

$$x_B = \frac{40}{50}$$

$$\therefore P_T = P_A^0 x_A + P_B^0 x_B$$

$$\Rightarrow 100 \times \frac{10}{50} + 150 \times \frac{40}{50} = 140 \text{ mm of Hg}$$

$$\text{Mole fraction of 'A' in vapour phase} = \frac{100 \times \frac{10}{50}}{140} = \frac{1}{7}$$

$$\therefore \text{Mole fraction of 'B' in vapour phase} = \frac{6}{7}$$

## PART-2 : CHEMISTRY

### SECTION-II (i)

3. **Ans ( 91.00 )**

$$Kt = \ln \left( \frac{V_\infty - V_0}{V_\infty - V_t} \right)$$

$$10^{-2} \times 230.3 = 2.303 \log \left( \frac{100 - 10}{100 - V_t} \right) \Rightarrow V_t = 91 \text{ ml}$$

4. **Ans ( 140.00 )**

$$r = K_1 [CH_3COO(CH_3)] \text{ where } K_1 = K [H^+][H_2O]$$

$$K_1 = (1.8 \times 10^{-3}) \times 0.1 \times \frac{1000}{18} = 10^{-2}$$

$$t = \frac{1}{K} \ln \frac{1}{0.25} = 140 \text{ sec}$$

## PART-2 : CHEMISTRY

### SECTION-II (ii)

8. **Ans ( 4 )**

$$[\sigma(g)1s]^2, [\sigma(u)1s]^2, [\sigma(g)2s]^2, [\sigma^*(g)1s]^2,$$

$$[\pi(u)2p_x]^1, [\pi(u)2p_y]^1$$

$$N_2 \Rightarrow [14 \text{ electrons}] = [\sigma(g)1s]^2, [\sigma(u)1s]^2,$$

$$[\sigma(g)2s]^2, [\sigma^*(u)2s]^2, [\sigma(g)2p_z]^2,$$

$$[\pi(u)2p_z]^2 = [\pi(u)2p_y]^2$$

$$C_2 \Rightarrow [12 \text{ electrons}] \Rightarrow [\sigma(g)1s]^2, [\sigma^*(u)1s]^2$$

$$[\sigma(g)2s]^2, [\sigma^*(u)2s]^2, [\pi(u)2p_x]^2 = [\pi(u)2p_y]^2$$

$$N_2^{2+} \Rightarrow [12 \text{ electrons}] [\sigma(g)1s]^2, [\sigma^*(u)1s]^2$$

$$[\sigma(g)2s]^2, [\sigma^*(u)2s]^2, [\pi(u)2p_x]^2 = [\pi(u)2p_y]^2$$

$$O_2 \Rightarrow [16 \text{ electrons}] \Rightarrow [\sigma(g)1s]^2, [\sigma^*(u)1s]^2,$$

$$[\sigma(g)2s]^2, [\sigma^*(u)2s]^2, [\sigma(g)2p_z]^2, [\pi(u)2p_x]^2$$

$$= [\pi(u)2p_y]^2, [\pi^*(g)2p_x]^1 = [\pi^*(u)2p_y]^1$$

$$N_2^{2-} \Rightarrow [16 \text{ electrons}] [\sigma(g)1s]^2, [\sigma^*(u)1s]^2,$$

$$[\sigma(g)2s]^2, [\sigma^*(u)2s]^2, [\sigma(g)2p_z]^2, [\pi(u)2p_x]^2$$

$$= [\pi(u)2p_y]^2, [\pi^*(g)2p_x]^1 = [\pi^*(u)2p_y]^1$$

$$O_2^{2-} \Rightarrow [18 \text{ electrons}] [\sigma(g)1s]^2, [\sigma^*(u)1s]^2,$$

$$[\sigma(g)2s]^2, [\sigma^*(u)2s]^2, [\sigma(g)2p_z]^2, [\pi(u)2p_x]^2$$

$$= [\pi(u)2p_y]^2, [\pi^*(g)2p_x]^2 = [\pi^*(u)2p_y]^2$$

## PART-3 : MATHEMATICS

### SECTION-I (i)

1. **Ans ( D )**

$$\int \left( \frac{2e^{\tan^{-1}x+x^2}}{1} \cdot \frac{(x)}{11} + x^2 \left( \frac{2x^3+2x+1}{x^2+1} \right) e^{\tan^{-1}x+x^2} \right) dx$$

$$2 \left[ e^{\tan^{-1}x+x^2} \left( \frac{x^2}{2} \right) - \int e^{\tan^{-1}x+x^2} \left( \frac{2x^3+2x+1}{x^2+1} \right) \frac{x^2}{2} dx \right]$$

$$+ \int x^2 \frac{(2x^3+2x+1)}{x^2+1} e^{\tan^{-1}x+x^2} dx$$

$$= x^2 e^{\tan^{-1}x+x^2} + C$$

2. **Ans (D)**

$$f'(x) = 2\alpha x - 12x^2 - \beta$$

$$\log_2 a + 2\log_2 a = \frac{\alpha}{6} \text{ (maxima \& minima are}$$

roots of  $f'(x) = 0$ )

$$\log_2 a = \frac{\alpha}{18} \quad \dots\dots (i)$$

$$f''(x) = 2\alpha - 24x = 0 \Rightarrow x = \frac{\alpha}{12}$$

$$\Rightarrow \frac{3}{4} = \frac{\alpha}{12} \text{ (Point of intersection)}$$

$$\alpha = 9$$

$$\Rightarrow \log_2 a = \frac{9}{18} \text{ \{from (i)\}}$$

$$\Rightarrow a = \sqrt{2}$$

3. **Ans (C)**

Let a point on  $y^3 = x^4$  be  $(t^3, t^4)$

$$3y^2 y' = 4x^3$$

$$\Rightarrow y' = \frac{4x^3}{3y^2} \Rightarrow y' = \frac{4}{3}t$$

Equation of tangent is

$$y - t^4 = \frac{4t}{3}(x - t^3)$$

$$\because \text{ it is a normal to } x^2 + y^2 - 2x = 0$$

$$\because \text{ it must pass through } (1, 0)$$

$$\Rightarrow -\frac{3}{4}t^3 = 1 - t^3 \Rightarrow \frac{t^3}{4} = 1$$

$$\Rightarrow t^3 = 4$$

$$\text{Now } m = \frac{4t}{3}$$

$$\Rightarrow \left(\frac{3m}{4}\right)^3 = t^3 = 4$$

4. **Ans (C)**

Consider  $f(x) = \sin^{-1}x$

by LMVT in  $[a, b]$

$$\frac{\sin^{-1}b - \sin^{-1}a}{b - a} = \frac{1}{\sqrt{1 - c^2}}, \text{ for atleast one}$$

$$c \in (a, b)$$

$$\sin^{-1}b - \sin^{-1}a = \frac{b - a}{\sqrt{1 - c^2}}$$

$$\Rightarrow b\sqrt{1 - a^2} - a\sqrt{1 - b^2} = \sin\left(\frac{b - a}{\sqrt{1 - c^2}}\right)$$

**PART-3 : MATHEMATICS**

**SECTION-I (ii)**

5. **Ans (A,C)**

$$f(\alpha) = \frac{1}{3}; f(f(\beta)) = \frac{1}{3}; f\left(\frac{1}{2}\right) = \frac{1}{2}$$

$$\Rightarrow f(\beta) = \alpha$$

$$\text{Let } \beta > \alpha$$

$$\Rightarrow \beta > f(\beta)$$

$$\Rightarrow \frac{f(\beta)}{\beta} < 1$$

$$\Rightarrow \alpha < \beta < \frac{1}{2}$$

$$\text{But } f(\alpha) < f\left(\frac{1}{2}\right)$$

$$\Rightarrow f \text{ is increasing}$$

$$\text{similarly } \beta < \alpha \Rightarrow f \text{ is decreasing}$$

$$\frac{f\left(\frac{1}{4}\right) - f(0)}{1/4} < \frac{f\left(\frac{1}{2}\right) - f\left(\frac{1}{4}\right)}{1/4}$$

$$\Rightarrow 2f\left(\frac{1}{4}\right) < f(0) + f\left(\frac{1}{2}\right)$$

6. **Ans (A,B)**

$$f(x) = e^{e^x} - \tan^{-1}x - 3$$

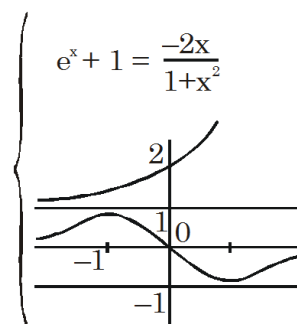
$$f'(x) = e^{e^x} \cdot e^x - \frac{1}{1 + x^2}$$

$$f'(x) = \frac{e^{e^x+x}(1 + x^2) - 1}{1 + x^2}$$

$$\text{Let } g(x) = e^{e^x+x}(1 + x^2) - 1$$

$$g'(x) = e^{e^x+x} \cdot \{e^x + 1\} \{1 + x^2\} + e^{e^x+x} \cdot 2x$$

$$g'(x) = e^{e^x+x} \{(e^x + 1)(1 + x^2) + 2x\}$$



$$\text{Now } (e^x + 1)(1 + x^2) + 2x > 0$$

$$\therefore g'(x) > 0 \forall x \in \mathbb{R}$$

$$\lim_{x \rightarrow -\infty} g(x) = -1 \text{ and } g(0) > 0$$

$$\therefore f(x) \text{ has negative minima}$$

$$\lim_{x \rightarrow -\infty} f(x) < 0 \text{ and } f(0) < 0$$

$$f(x) = 0 \text{ has positive real root}$$

7. **Ans (D)**

All the three roots of the equation  $x^3 - 3x + 1 = 0$  are real and irrational

So,  $n! x_i$  can't be an integer for any  $n$

$$\text{So, } \left| \lim_{n \rightarrow \infty} \cos(\pi n! x_i) \right| < 1$$

$$\text{So, } \lim_{m \rightarrow \infty} \left( \lim_{n \rightarrow \infty} \cos(\pi n! x_i) \right)^m = 0, \text{ for every } i = 1, 2, 3$$

8. **Ans (B,C)**

$$\text{Let } G(x) = e^x \cos x f(x)$$

$$G'(x) = \cos x f'(x) + (\cos x - \sin x) f(x) \leq 0$$

$G(x)$  is a non increasing function

$$G(\pi/2) \geq G(x), \forall x \geq \pi/2$$

$$G(6) \leq 0$$

$$e^6 \cos 6 f(6) \leq 0$$

$$\Rightarrow f(6) \leq 0 \text{ but } f(x) \geq 0$$

$$\Rightarrow f(6) = 0$$

9. **Ans (A,C)**

$$f(x) = g\left(\frac{1}{x^5}\right) e^{-\frac{1}{x^6}}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{g\left(\frac{1}{x^5}\right)}{e^{\frac{1}{x^6}}}$$

$$= \lim_{t \rightarrow \infty} \frac{g(t^5)}{e^{t^6}}$$

$$f'(x) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{g\left(\frac{1}{x^5}\right) e^{-\frac{1}{x^6}}}{x} = \lim_{x \rightarrow 0} \frac{g\left(\frac{1}{x^5}\right)}{x e^{\frac{1}{x^6}}}$$

$$= \lim_{t \rightarrow \infty} \frac{t g(t^5)}{e^{t^6}} = 0$$

$\Rightarrow f(x)$  is continuous and differentiable at  $x = 0$

10. **Ans (A,C,D)**

$$f(|x|) = \{|x| - 2, 0 \leq |x| \leq 3 \Rightarrow f(|x|) = \begin{cases} -x - 2, -3 \leq x \leq 0 \\ x - 2, 0 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(|x|)] = \begin{cases} [-x] - 2, -3 \leq x \leq 0 \\ [x] - 2, 0 < x \leq 3 \end{cases}$$

$$\text{Also } |f(x)| = \begin{cases} 2, -3 \leq x \leq 0 \\ 2 - x, 0 < x \leq 2 \\ x - 2, 2 < x \leq 3 \end{cases}$$

$$\Rightarrow [|f(x)|] = \begin{cases} 2, -3 \leq x \leq 0 \\ 2 + [-x], 0 < x \leq 2 \\ [x] - 2, 2 < x \leq 3 \end{cases}$$

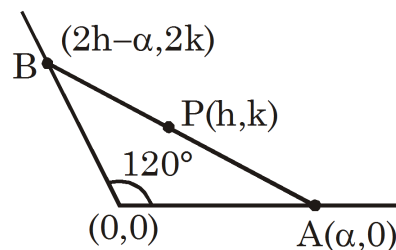
$$\text{So } [f|x|] + [|f(x)|] = \begin{cases} [-x], -3 \leq x \leq 0 \\ [x] + [-x], 0 < x \leq 2 \\ 2[x] - 4, 2 < x \leq 3 \end{cases}$$

Now options can be checked directly without further simplification.

**PART-3 : MATHEMATICS**

**SECTION-II (i)**

1. **Ans (10.00)**



$$\text{Let } (\alpha, 0) \Rightarrow B(2h - \alpha, 2k)$$

$$\frac{2k}{2h - \alpha} = -\sqrt{3} \Rightarrow \alpha = \frac{2k + 2\sqrt{3}h}{\sqrt{3}}$$

$$4k^2 + (2h - 2\alpha)^2 = 100$$

$$3x^2 + 7y^2 + 4\sqrt{3}xy = 75$$

$$\text{Let } x = r \cos \theta, y = r \sin \theta$$

$$\Rightarrow r^2 = \frac{75}{3 \cos^2 \theta + 7 \sin^2 \theta + 4\sqrt{3} \sin \theta \cos \theta}$$

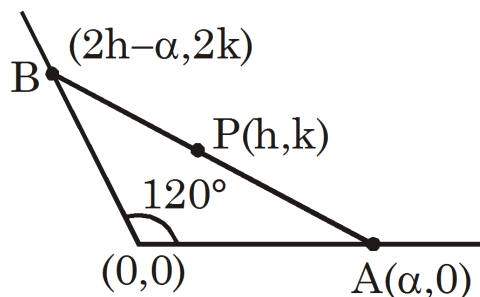
$$r^2 = \frac{75}{3 + 4 \sin^2 \theta + 2\sqrt{3} \sin 2\theta}$$

$$r^2 = \frac{75}{5 + (2\sqrt{3} \sin 2\theta - 2 \cos 2\theta)}$$

$$r_{\max}^2 = \frac{75}{5 + (-4)} = 75$$

$$\Rightarrow r_{\max} = 5\sqrt{3}$$

2. Ans ( 15.00 )



Let  $(\alpha, 0) \Rightarrow B(2h - \alpha, 2k)$

$$\frac{2k}{2h - \alpha} = -\sqrt{3} \Rightarrow \alpha = \frac{2k + 2\sqrt{3}h}{\sqrt{3}}$$

$$4k^2 + (2h - 2\alpha)^2 = 100$$

$$3x^2 + 7y^2 + 4\sqrt{3}xy = 75$$

Let  $x = r\cos\theta$ ,  $y = r\sin\theta$

$$\Rightarrow r^2 = \frac{75}{3\cos^2\theta + 7\sin^2\theta + 4\sqrt{3}\sin\theta\cos\theta}$$

$$r^2 = \frac{75}{3 + 4\sin^2\theta + 2\sqrt{3}\sin 2\theta}$$

$$r^2 = \frac{75}{5 + (2\sqrt{3}\sin 2\theta - 2\cos 2\theta)}$$

$$r_{\max}^2 = \frac{75}{5 + (-4)} = 75$$

$$\Rightarrow r_{\max} = 5\sqrt{3}$$

3. Ans ( 8.00 )

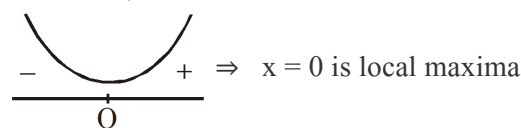
$$f(x) = \sin^{-1}(\sin e^x)$$

$$f'(x) = \frac{1}{\sqrt{1 - \sin^2(e^x)}} \cdot \cos(e^x) \cdot e^x$$

$$f'(x) = \frac{\cos(e^x)}{|\cos e^x|} \cdot e^x \Rightarrow \text{non-monotonic}$$

$$g(x) = \sin^{-1}\left(\frac{3^x + 3^{-x}}{4}\right) \Rightarrow \text{even function}$$

$$g'(x) = \frac{1}{\sqrt{1 - \left(\frac{3^x + 3^{-x}}{4}\right)^2}} \cdot \frac{(3^x - 3^{-x})\ln 3}{4}$$



Domain of  $f(x)$  is  $\mathbb{R}$

$$\text{Domain of } g(x) - 1 \leq \frac{3^x + 3^{-x}}{4} \leq 1$$

$$(3^x)^2 - 4(3^x) + 1 \leq 0$$

$$3^x \in [2 - \sqrt{3}, 2 + \sqrt{3}]$$

$$\Rightarrow x \in [-\log_3(2 + \sqrt{3}), \log_3(2 + \sqrt{3})]$$

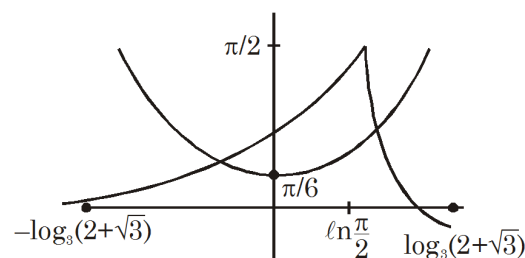
domain of  $h(x)$  is domain of  $g(x)$

$\Rightarrow$  D is incorrect

number of solution of  $h(x) = 0$

$$\Rightarrow f(x) = g(x)$$

$$\sin^{-1}(\sin e^x) \Rightarrow \begin{cases} e^x & x \leq \ln \frac{\pi}{2} \\ \pi - e^x & \ln \frac{\pi}{2} < x \leq \ln \frac{3\pi}{2} \\ e^x - 2\pi & \ln \frac{3\pi}{2} < x \leq \ln \frac{5\pi}{2} \end{cases}$$



$$\text{Range of } g(x) \Rightarrow \left[\frac{\pi}{6}, \frac{\pi}{2}\right]$$

$\Rightarrow$  2 solutions.

4. Ans (2.00)

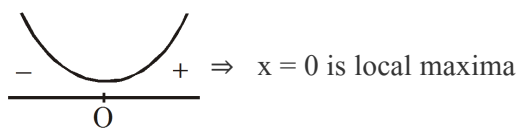
$$f(x) = \sin^{-1}(\sin e^x)$$

$$f'(x) = \frac{1}{\sqrt{1 - \sin^2(e^x)}} \cdot \cos(e^x) \cdot e^x$$

$$f'(x) = \frac{\cos(e^x)}{|\cos e^x|} \cdot e^x \Rightarrow \text{non-monotonic}$$

$$g(x) = \sin^{-1}\left(\frac{3^x + 3^{-x}}{4}\right) \Rightarrow \text{even function}$$

$$g'(x) = \frac{1}{\sqrt{1 - \left(\frac{3^x + 3^{-x}}{4}\right)^2}} \cdot \frac{(3^x - 3^{-x}) \ln 3}{4}$$



Domain of  $f(x)$  is  $\mathbb{R}$

$$\text{Domain of } g(x) - 1 \leq \frac{3^x + 3^{-x}}{4} \leq 1$$

$$(3^x)^2 - 4(3^x) + 1 \leq 0$$

$$3^x \in [2 - \sqrt{3}, 2 + \sqrt{3}]$$

$$\Rightarrow x \in [-\log_3(2 + \sqrt{3}), \log_3(2 + \sqrt{3})]$$

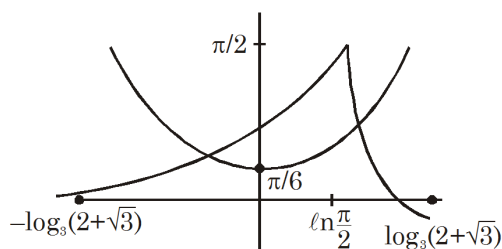
domain of  $h(x)$  is domain of  $g(x)$

$\Rightarrow$  D is incorrect

number of solution of  $h(x) = 0$

$$\Rightarrow f(x) = g(x)$$

$$\sin^{-1}(\sin e^x) \Rightarrow \begin{cases} e^x & x \leq \ln \frac{\pi}{2} \\ \pi - e^x & \ln \frac{\pi}{2} < x \leq \ln \frac{3\pi}{2} \\ e^x - 2\pi & \ln \frac{3\pi}{2} < x \leq \ln \frac{5\pi}{2} \end{cases}$$



$$\text{Range of } g(x) \Rightarrow \left[\frac{\pi}{6}, \frac{\pi}{2}\right]$$

$\Rightarrow$  2 solutions.

5. Ans (0.50)

$$2f(x) \cdot f'(x) - 2f'(x) - \frac{2}{2\sqrt{x}} = 0$$

$$2(f(x) - 1)f'(x) = \frac{1}{2\sqrt{x}}$$

integrating both the sides

$$(f(x) - 1)^2 - \sqrt{x} + C$$

$$\therefore f(0) = 2c = 1$$

$$\Rightarrow (f(x) - 1)^2 = \sqrt{x} + 1$$

$$\Rightarrow f(x) = 1 + \sqrt{\sqrt{x} + 1}$$

$$\text{Now, } \lim_{x \rightarrow 0^+} \frac{f(x) - 2}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \frac{\sqrt{\sqrt{x} + 1} - 1}{\sqrt{x}} \\ = \lim_{x \rightarrow 0^+} \frac{\sqrt{x + 1} - 1}{\sqrt{x}(\sqrt{\sqrt{x} + 1} + 1)} = \frac{1}{2}$$

6. Ans (96.00)

$$\therefore g'(y) = \frac{1}{f'(x)} \Rightarrow g'(4) = \frac{1}{f'(64)}$$

$$(\because f(64) = 4)$$

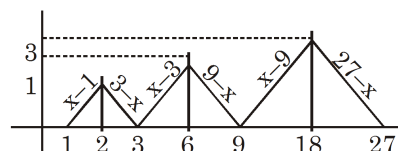
$$= \frac{1}{\frac{1}{96}} = 96$$

## PART-3 : MATHEMATICS

### SECTION-II (ii)

7. Ans (429)

$$f(x) = \begin{cases} x - 3^n; & 3^n \leq x \leq 2 \times 3^n \\ 3^{(n+1)} - x; & 2 \times 3^n < x \leq 3^{(n+1)} \end{cases}; n \in \mathbb{N} \cup \{0\}$$



$$\text{Since } 2 \times 3^6 < 2001 < 3^7$$

$$\Rightarrow f(2001) = 3^7 - 2001 = 186$$

$$\text{Now, } f(x) \leq 81, \text{ when } x \leq 3^5$$

$$\text{And } 0 \leq f(x) < 3^5 = 243$$

$$\text{when } 3^5 \leq x \leq 2 \times 3^5$$

$$\text{So, } 186 = x - 3^5$$

$$\Rightarrow x = 243 + 186 = 429$$

8. **Ans ( 2 )**

$$(13\ell + 19m - 32) + n(11\ell + 17m - 28) = 0$$

$$\text{Let } (13x + 19y - 32) + n(11x + 17y - 28) = 0$$

Equation is family of line passes through (1, 1)

$$p + q = 2$$

 9. **Ans ( 4 )**

$$\text{Let } g(x) = \sin^{-1} |\sin x| + \cos^{-1} (\cos x)$$

$$= \begin{cases} 2x & 0 \leq x \leq \frac{\pi}{2} \\ \pi & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ 4\pi - 2x & \frac{3\pi}{2} < x \leq 2\pi \end{cases}$$

So  $f(x)$  will be const. in  $\left[ \frac{2n\pi}{k} + \frac{\pi}{2k}, \frac{2n\pi}{k} + \frac{3\pi}{2k} \right]$

$$\frac{\pi}{4} = \frac{3\pi}{2k} - \frac{\pi}{2k} \Rightarrow k = 4$$

**JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS & TNAS]**

**ANSWER KEY**

**PAPER-2**

## PART-1 : PHYSICS

|            |    |    |    |    |     |    |    |    |      |    |    |
|------------|----|----|----|----|-----|----|----|----|------|----|----|
| SECTION-I  | Q. | 1  | 2  | 3  | 4   | 5  | 6  | 7  | 8    | 9  | 10 |
|            | A. | B  | B  | A  | B   | D  | C  | D  | A    | B  | B  |
|            | Q. | 11 | 12 | 13 | 14  | 15 | 16 | 17 | 18   | 19 | 20 |
|            | A. | A  | B  | A  | C   | A  | D  | C  | C    | A  | D  |
| SECTION-II | Q. | 1  | 2  | 3  | 4   | 5  | 6  | 7  | 8    | 9  | 10 |
|            | A. | 11 | 2  | 0  | 100 | 2  | 3  | 2  | 1500 | 5  | 2  |

**PART-2 : CHEMISTRY**

|            |    |    |    |    |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|----|----|----|----|
| SECTION-I  | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|            | A. | D  | C  | A  | D  | C  | D  | C  | C  | C  | B  |
|            | Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|            | A. | A  | A  | B  | C  | D  | C  | B  | D  | A  | B  |
| SECTION-II | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|            | A. | 6  | 22 | 3  | 3  | 7  | 3  | 5  | 64 | 1  | 2  |

### PART-3 : MATHEMATICS

|            |    |    |    |    |     |    |    |     |    |    |    |
|------------|----|----|----|----|-----|----|----|-----|----|----|----|
| SECTION-I  | Q. | 1  | 2  | 3  | 4   | 5  | 6  | 7   | 8  | 9  | 10 |
|            | A. | D  | B  | A  | A   | B  | D  | C   | B  | A  | C  |
|            | Q. | 11 | 12 | 13 | 14  | 15 | 16 | 17  | 18 | 19 | 20 |
|            | A. | D  | D  | A  | A   | A  | D  | D   | A  | B  | C  |
| SECTION-II | Q. | 1  | 2  | 3  | 4   | 5  | 6  | 7   | 8  | 9  | 10 |
|            | A. | 6  | 5  | 4  | 512 | 4  | 5  | 189 | 17 | 33 | 2  |

## HINT – SHEET

## PART-1 : PHYSICS

## SECTION-I

1. **Ans ( B )**

$$\text{Current in EC} = \frac{40}{20} = 2\text{A}$$

Current at C =  $2 + 2 = 4A$

$$\text{Equivalent resistance of CD} = \frac{5 \times 5}{10} = 2.5\Omega$$

Potential drop in CD =  $4 \times 2.5 = 10V$

$$100 - V_{EC} + V_{CD} + V_{DF}$$

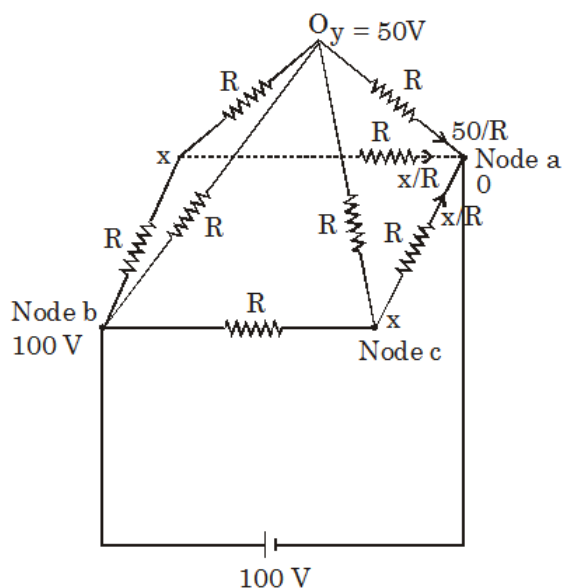
$$V_{DE} = 100 - 40 - 10 = 50V$$

Current in wire DG = 2A

Current in resistance DF = 2A

$$R \times 2 = 50 \text{ or } R = 25\Omega$$

2. Ans ( B )



Step 2: By symmetry  $100 - y = y - 0$  or  $y = 50V$

Step 3: Now apply KCL at node c

$$\frac{x-100}{R} + \frac{x+50}{R} + \frac{x-0}{R} = 0$$

or  $x = 50V$

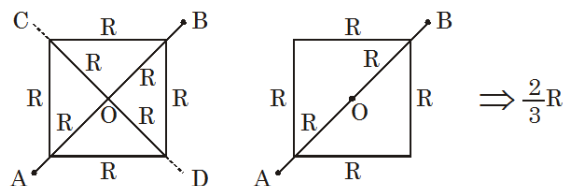
Step 4: If  $R_{eq}$  is equivalent resistance of this circuit, we have

$$\frac{100}{R_{eq}} = \frac{x}{R} + \frac{50}{R} + \frac{50}{12} \text{ or } R_{eq} = \frac{2}{3}R$$

**Alternative method:**

Making 2D projection

There is perpendicular axis of symmetry, i.e., the points COD will be equipotential then the circuit will be



3. **Ans (A)**

Resistance of each part =  $\frac{R}{2n}$ . For 'n' such parts connected in series, equivalent resistances, say  $R_1 = n$

$$\left[ \frac{R}{2n} \right] = \frac{R}{2}$$

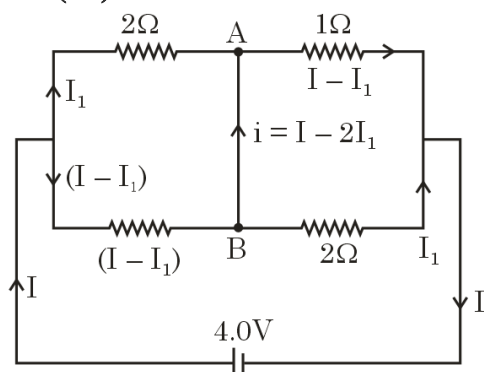
Similarly, equivalent resistance say  $R_2$  for another set of n identical respectively in parallel would be

$$\frac{1}{n} \left( \frac{R}{2n} \right) = \frac{R}{2n^2}$$

For getting maximum of  $R_1$  &  $R_2$ , they should be connected in series & hence,

$$R_{eq} = R_1 + R_2 = \frac{R}{2} \left( 1 + \frac{1}{n^2} \right)$$

4. **Ans (B)**



Using KVL, we get

$$2I_1 - (I - I_1) = 0 \text{ or } I = 3I_1$$

$$\text{and } (I - I_1) + 2I_1 = 4$$

Solving, we get  $I_1 = 1A$  and  $I = 3A$

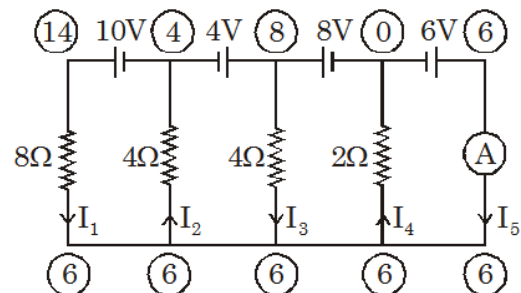
Therefore, current through AB =  $I - 2I_1 = (3 - 2)A = 1A$

5. **Ans (D)**

$$k = \frac{E_1}{100} = \frac{2}{100} = 0.02 \text{ V cm}^{-1}$$

$$E_2 = k(75) = 0.02 \times 75 = 1.5 \text{ V}$$

6. **Ans (C)**



The current through ammeter

$$\begin{aligned} I &= I_4 - I_3 + I_2 - I_1 \\ &= \left( \frac{6-0}{2} \right) - \left( \frac{8-6}{4} \right) + \frac{(6-4)}{4} - \frac{(14-6)}{8} \\ &= 3 - \frac{1}{2} + \frac{1}{2} - 1 = 3 - 1 = 2A \end{aligned}$$

7. **Ans (D)**

Let  $\rho$  = density of sphere,  $R$  = radius of sphere,  $r$  = radius of the spherical cavities.

$$\text{Mass of the complete sphere} = \frac{4}{3}\pi R^3 \rho = M$$

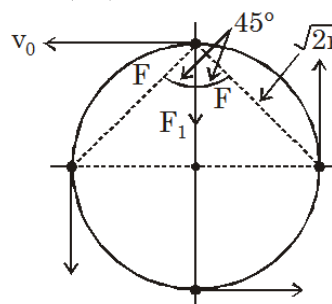
$$\text{Mass of the removed sphere} = \frac{4}{3}\pi r^3 \rho = m$$

$$\text{Here } m = \frac{Mr^3}{R^3} = \frac{M(1)^3}{(4)^3} = \frac{M}{64}$$

$$\text{Now } \vec{I}_R = \vec{I} + \vec{I}_P + \vec{I}_Q$$

$$\text{Here } I = 0, \text{ also } \vec{I}_P = -\vec{I}_Q \Rightarrow I_R = 0$$

8. **Ans (A)**



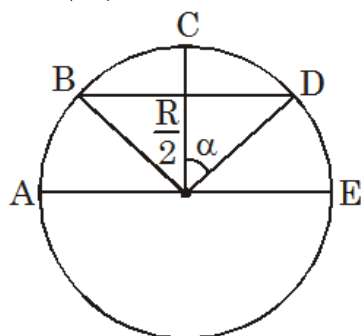
Centripetal force = net gravitational force

$$\Rightarrow mv_0^2 = 2F \cos 45^\circ + F_1 = \frac{2Gm^2}{(\sqrt{2})^2 \sqrt{2}} + \frac{Gm^2}{4r^2}$$

$$\frac{mv_0^2}{r} = \frac{GM^2}{4r^2} [2\sqrt{2} + 1]$$

$$\Rightarrow v_0 = \left[ \frac{Gm(2\sqrt{2} + 1)}{4r} \right]^{\frac{1}{2}}$$

9. Ans (B)



$$\alpha = 60^\circ$$

Solid angle subtended by BCD is

$$\omega = 2\pi(1 - \cos \alpha) = \pi$$

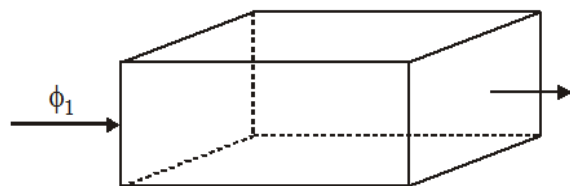
Solid angle subtended by ABDE is

$$\omega_{(ABCDE)} - \omega_{(BCD)} = 2\pi - \pi = \pi$$

Hence, flux through ABDE is

$$\phi = \frac{q}{\epsilon_0} \frac{\pi}{4\pi} = \frac{q}{4\epsilon_0}$$

10. Ans (B)



$$\phi_1 = -E_1 a^2 = -\alpha a^{1/2} a^2 = -\alpha a^{5/2}$$

$$\text{and } \phi_2 = E_2 a^2 = \alpha(2a)^{1/2} a^2 = \sqrt{2}\alpha a^{5/2}$$

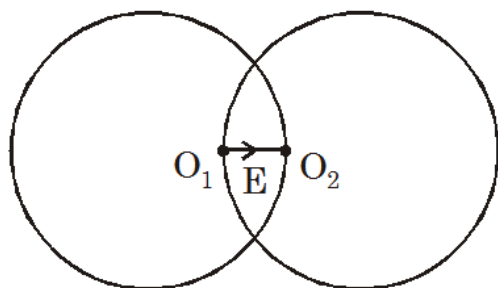
$$\phi_{\text{net}} = (\sqrt{2} - 1)\alpha a^{5/2}$$

Using the Gauss theorem, we get

$$\phi = \frac{q_{\text{in}}}{\epsilon_0}$$

$$\text{or } q_{\text{in}} = \phi \epsilon_0 = (\sqrt{2} - 1)\epsilon_0 \alpha a^{5/2}$$

11. Ans (A)



$$d = R$$

Electric field from  $O_1$  through  $O_2$  is uniform and is  $E = \rho R / 3\epsilon_0$

So potential difference is

$$\Delta V = ER = \frac{\rho R^2}{3\epsilon_n} = \frac{\rho d^2}{3\epsilon_0}$$

13. Ans (A)

Charge can be considered as located at a distance  $R$  from the center. Total charge is

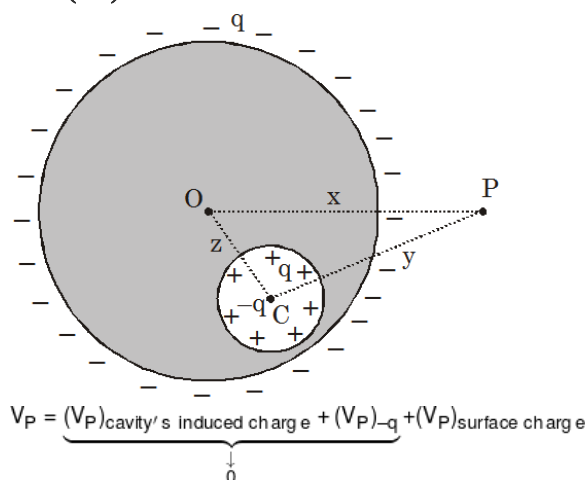
$$(Q - 2Q + 3Q) = 2Q$$

$$\text{Hence } V = \frac{1}{4\pi\epsilon_0} \frac{2Q}{R} = \frac{Q}{2\pi\epsilon_0 R}$$

17. Ans (C)

$$T = 2\pi \sqrt{\frac{\ell}{g \left(1 + \frac{\ell}{R}\right)}}$$

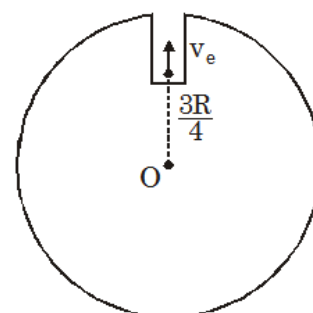
18. Ans (C)



$$V_P = \underbrace{(V_P)_{\text{cavity's induced charge}} + (V_P)_{-q}}_0 + (V_P)_{\text{surface charge}}$$

$$V_P = -\frac{kq}{x}$$

19. Ans (A)



COE

$$\frac{1}{2} m v_e^2 + m v = 0 \quad \dots (i)$$

$$v = \frac{-GM}{2R^3} (3R^2 - r^2)$$

$$v = \frac{-GM}{2R^3} \left( 3R^2 - \frac{9R^2}{16} \right)$$

$$v = \frac{-39GM}{32R}$$

$$\text{by equation (i)} \quad \frac{1}{2} m v_e^2 + \left( \frac{-39GMm}{32R} \right) = 0$$

$$v_e = \sqrt{\frac{39Gm}{16R}} = \frac{1}{4} \sqrt{\frac{39Gm}{R}}$$

20. **Ans ( D )**

For  $\theta < \theta_C$  (critical angle),  $\delta = r - i$

For  $\theta > \theta_C$ ,  $\delta = \pi - 2i$

## PART-1 : PHYSICS

### SECTION-II

9. **Ans ( 5 )**

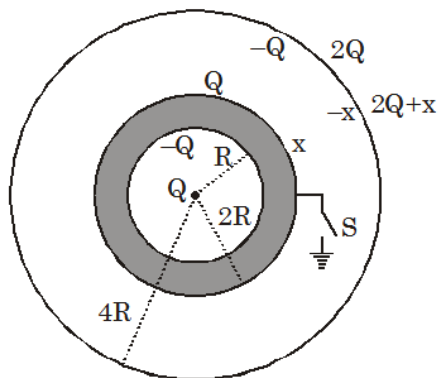
$$J = \sigma E$$

$$i/A = \sigma E \Rightarrow E = \frac{i}{\sigma A}$$

$$E \propto 1/A$$

$$\frac{E_1}{E_2} = \frac{A_2}{A_1} = \frac{10^2}{2^2} = 25$$

10. **Ans ( 2 )**



$$v = 0$$

$$\frac{kx}{2R} + \frac{k2Q}{4R}$$

$$\frac{x}{2} = -\frac{2Q}{4R}$$

$$x = -Q$$

charge flown into the earth = 2Q

## PART-2 : CHEMISTRY

### SECTION-I

5. **Ans ( C )**

Fact

14. **Ans ( C )**

$$P = P_A^0 x_A + P_B^0 x_B$$

$$600 = \left( \frac{3}{3+2} \right) P_A^0 + P_B^0 \left( \frac{2}{2+3} \right); 3P_A^0 + 2P_B^0$$

$$= 3000 \text{ \&}$$

$$630 = P_A^0 \left( \frac{4.5}{4.5+2+0.5} \right) + P_B^0 \left( \frac{2}{4.5+2+0.5} \right)$$

$$4.5 P_A^0 + 2P_B^0 = 4410$$

$$1.5 P_A^0 = 1410; P_A^0 = 940 \text{ \& } P_B^0 = 90$$

16. **Ans ( C )**

$$X_{N_2} = 0.8, X_{O_2} = 0.2 \text{ in air}$$

$$\therefore P_{N_2} = 0.8 \text{ atm}, P_{O_2} = 0.2 \text{ atm}$$

$P = K_H X_{\text{gas}}$  where X is solubility of gas

$$0.8 = 2 \times 10^4 X_{N_2} \dots (1)$$

$$0.2 = 10^4 X_{O_2} \dots (2)$$

Ratio of solubility of  $N_2$  &  $O_2$  is 2 : 1

17. **Ans ( B )**

$$r = K [A]^2 [B]^0$$

$$10^{-2} = K(1)^2(1) \Rightarrow K = 10^{-2}$$

$$\text{finally, } r = 10^{-2} (1/2)^2 (1/2)^0 = 2.5 \times 10^{-3}$$

20. **Ans ( B )**

$$X \rightarrow 7/8$$

$$Y \rightarrow \frac{1}{2} \times 6$$

$$Z \rightarrow \frac{1}{8}$$

## PART-2 : CHEMISTRY

### SECTION-II

8. **Ans ( 64 )**

For a 'n' the order reaction,

$$t_{1/2} \propto \frac{1}{a^{n-1}} \text{ where } a = \text{initial conc.}$$

$$\left( \frac{5 \times 10^{-3}}{25 \times 10^{-4}} \right)^{n-1} = \frac{8}{1}$$

$$\frac{t_{1/2}}{1} = \left( \frac{5 \times 10^{-3}}{1.25 \times 10^{-3}} \right)^{n-1}$$

$$t_{1/2} = 64 \text{ hr}$$

9. **Ans ( 1 )**

$$\frac{y}{x} = \frac{1}{\text{antilog}(0.3)} = \frac{1}{2} = 0.50$$

10. **Ans ( 2 )**

$$a - 2r = 1.2 \text{ \AA} \text{ and } \sqrt{3} a = 4r$$

$$a - 2 \times \frac{\sqrt{3}}{4} a = 1.2 \text{ \AA}$$

$$a \left( 1 - \frac{\sqrt{3}}{2} \right) = 1.2 \text{ \AA}$$

$$a = 8 \text{ \AA}$$

## PART-3 : MATHEMATICS

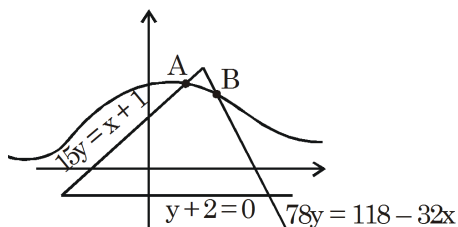
### SECTION-I

1. **Ans (D)**

$$\begin{aligned} \sum_{r=0}^n \tan^{-1} (1+r+r^2) \\ &= \sum_{r=0}^n \frac{\pi}{2} - \tan^{-1} \frac{1}{1+r+r^2} \\ &= \frac{n\pi}{2} - \sum_{r=0}^n \tan^{-1} \frac{(r+1)-r}{1+r(r+1)} \\ &= \frac{n\pi}{2} - \sum_{r=0}^n (\tan^{-1} (r+1) - \tan^{-1} r) \end{aligned}$$

2. **Ans (B)**

P lies on  $y = \frac{1}{1+x^2}$



$A\left(2, \frac{1}{5}\right), B\left(5, \frac{1}{26}\right)$

hence  $a \in [2, 5]$

4. **Ans (A)**

take  $2x + \frac{1}{x^2} = t$

5. **Ans (B)**

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0)$$

$$\Rightarrow b = a = a \quad \dots\dots(1)$$

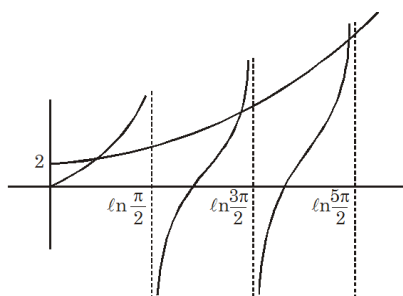
$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{f(0-h) - f(0)}{h}$$

$$1 = a \quad \dots\dots(2)$$

$$\therefore a = 1 = b$$

6. **Ans (D)**

$$\tan e^x = e^x + e^{-x}$$



there are infinite solutions.

7. **Ans (C)**

Given slope of normal at  $(4, 6) = \tan \frac{2\pi}{3}$

$$-\frac{1}{\left(\frac{dy}{dx}\right)_{(4,6)}} = -\sqrt{3}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(4,6)} = \frac{1}{\sqrt{3}}$$

8. **Ans (B)**

$$\lim_{x \rightarrow 0} \frac{5 \sin x + x \cos x}{2 \tan x - x^2}$$

$$\lim_{x \rightarrow 0} \frac{5 \frac{\sin x}{x} + \cos x}{\frac{2 \tan x}{x} - x} = \frac{5+1}{2-0} = 3$$

9. **Ans (A)**

$$\int [(1-2x) \cos x + (3+2x) \sin x] dx$$

$$\int ((\cos x + 3 \sin x) + 2x (\sin x - \cos x)) dx$$

$$\sin x - 3 \cos x + 2x(-\sin x - \cos x) - \int 2(-\cos x - \sin x) dx$$

$$\sin x(3-2x) + \cos x(-5-2x)$$

10. **Ans (C)**

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

$$3ax^2 + 2bx + c > 0 \quad \forall x \in \mathbb{R}$$

$$\text{Let } g(x) = 3ax^2 + 2bx + c > 0 \quad \forall x \in \mathbb{R}$$

$$g(1)g(0) > 0$$

$$(3a + 2b + c)(c) > 0$$

total possible cases 21

11. **Ans (D)**

$$f(x) = 3 - x^2, 1 \leq x \leq 4$$

$$f(2x) = 3 - 4x^2, 1 \leq 2x \leq 4$$

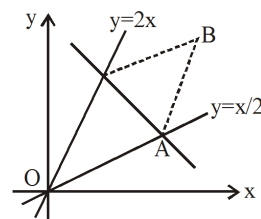
$$\text{for } \log_e(f(2x)), 3 - 4x^2 > 0 \text{ \& } 2x \in [1, 4]$$

$$x \in \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

12. **Ans (D)**

$$A\left(\frac{1}{6}, \frac{1}{12}\right)$$

$$C\left(\frac{1}{9}, \frac{2}{9}\right)$$

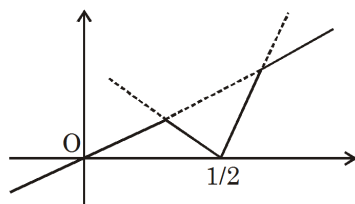


$$\Delta = 2\text{Ar. } (\Delta AOC) = 2 \times \frac{1}{2} \left| \frac{1}{6} \times \frac{2}{9} - \frac{1}{9} \times \frac{1}{12} \right|$$

$$= \frac{1}{36} \text{ sq. units}$$

13. Ans (A)

$$|x + |2x - 1|| - |x - |2x - 1|| = 2 \min(x, |2x - 1|)$$



$$|2x - 1| = x$$

$$\therefore 2x - 1 = \pm x$$

$$x = 1, \frac{1}{3}$$

$$\text{at } x = \frac{1}{3}$$

14. Ans (A)

$$\begin{aligned} & \int (6x^2 + 5x + 4)(x^2 + x + 1)^6 \cdot x^{27} dx \\ &= \int (x^2 + x + 1)^6 \cdot x^{24} (6x^5 + 5x^4 + 4x^3) dx \\ &= \int (x^6 + x^5 + x^4)^6 (6x^5 + 5x^4 + 4x^3) dx \\ &= \int t^6 dt = \frac{t^7}{7} + C = \frac{(x^6 + x^5 + x^4)^7}{7} + C \\ &= \frac{x^{28}(x^2 + x + 1)^7}{7} + C \end{aligned}$$

15. Ans (A)

$$\begin{aligned} L &= \lim_{x \rightarrow \infty} \frac{x^{\frac{1}{3}}((x+1) - (x-1)) \left( (x+1)^{\frac{1}{3}} + (x-1)^{\frac{1}{3}} \right)}{(x+1)^{\frac{2}{3}} + (x^2-1)^{\frac{1}{3}} + (x-1)^{\frac{2}{3}}} \\ &= \lim_{x \rightarrow \infty} \frac{2 \left( (x^2+x)^{\frac{1}{3}} + (x^2-x)^{\frac{1}{3}} \right)}{(x+1)^{\frac{2}{3}} + (x^2-1)^{\frac{1}{3}} + (x-1)^{\frac{2}{3}}} \\ &= \frac{2(2)}{3} = \frac{4}{3} \end{aligned}$$

16. Ans (D)

$$f(x) = -x \ln x$$

$$f'(x) = -1 - \ln x$$

$$\therefore f'(x) > 0 \text{ if } x \in (0, 1/e)$$

$$f'(x) = 0 \text{ if } x = 1/e$$

$$f'(x) < 0 \text{ if } x > 1/e$$

$$f(1/e) = e^{-1}$$

$$\text{and } \lim_{x \rightarrow 0^+} f(x) = 0 \text{ and } \lim_{x \rightarrow 1^-} f(x) = 0$$

17. Ans (D)

$$-1 \leq x - 1 \leq 1, -1 \leq x - 3 \leq 1, x^2 \neq 2$$

$$\Rightarrow 0 \leq x \leq 2, 2 \leq x \leq 4, x \neq \pm \sqrt{2}$$

$$\therefore x = 2$$

$$\therefore \text{RHS} = \sin^{-1}(1) + \cos^{-1}(-1) + \tan^{-1}(-1)$$

$$= \frac{\pi}{2} + \pi - \frac{\pi}{4} = \frac{\pi}{2} + \frac{3\pi}{4} = \frac{5\pi}{4}$$

18. Ans (A)

$$2a + b = 3, a^2 + 2ab = -9$$

$$\Rightarrow a^2 + 2a(3 - 2a) = -9$$

$$\Rightarrow a^2 - 4a^2 + 6a + 9 = 0$$

$$\Rightarrow -3a^2 + 6a + 9 = 0$$

$$\Rightarrow a^2 - 2a - 3 = 0$$

$$\Rightarrow (a + 1)(a - 3) = 0$$

$$a = -1, 3$$

$$c = -a^2b$$

$$\therefore a = -1, b = 5;$$

$$a = 3, b = -3$$

$$\therefore c = -5 \text{ or } 27$$

19. Ans (B)

$$f(x) = \frac{9}{4}x^2 + \frac{4}{9}$$

$$f'(x) = \frac{9}{2}x$$

$$f''(x) = \frac{9}{2}$$

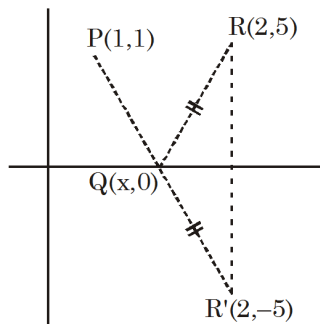
20. Ans (C)

$$\text{at } x = 1, 2$$

## PART-3 : MATHEMATICS

### SECTION-II

1. **Ans (6)**



$$f(x) = \sqrt{(x-1)^2 + (0-1)^2} + \sqrt{(x-2)^2 + (0-5)^2}$$

$$f(x) = PQ + QR$$

∴  $f(x)$  will be minimum if P, Q, R' are collinear.

$$\therefore f(x) = PR' = \sqrt{1+36} = \sqrt{37} = m$$

$$\Rightarrow [m] = 6$$

2. **Ans (5)**

$$f'(x) = 24x^{1/3} - 3x^{-2/3}$$

$$f'(x) = 0 \Rightarrow x = \frac{1}{8}$$

$$f(-1) = 18 + 9 + 1 = 28$$

$$f(8) = 18(2^3)^{4/3} - 9(2^3)^{1/3} + 1$$

$$= 18 \cdot 16 - 18 + 1 = 271$$

$$f\left(\frac{1}{8}\right) = 18 \cdot \frac{1}{16} - \frac{9}{2} + 1 = -\frac{19}{8}$$

$$\text{clearly, } a = 28, b = -\frac{19}{8}$$

$$a + 8b = 28 - 8 \cdot \frac{19}{8} = 28 - 19 = 9$$

3. **Ans (4)**

$$f(1) = 8, f(2) = 12, f'(1) = 5, f'(2) = 2$$

Now

$$L = \lim_{h \rightarrow 0} \frac{(4 \cosh - 3 \sinh) \cdot f'(3 \cosh + 4 \sinh - 2)}{(3e^h - 5 \sec h \tanh) \cdot f'(3e^h - 5 \sec h + 4)}$$

(using L' hospital rule)

$$= \frac{4f'(1)}{3f'(2)} = \frac{24}{6} = 4$$

4. **Ans (512)**

$$L = \lim_{x \rightarrow 0} \frac{\left(1 + ax + \frac{a^2 x^2}{2!} + \frac{a^3 x^3}{3!}\right) + \left(4x - \frac{16x^2}{2} + \frac{64x^3}{3}\right) + b}{x^3}$$

$$L = \lim_{x \rightarrow 0} \frac{(1+b) + (a+4)x + \left(\frac{a^2}{2} - 8\right)x^2 + \left(\frac{a^3}{3} + \frac{64}{3}\right)x^3}{x^3}$$

$$\Rightarrow a = -4, b = -1, L = \frac{a^3}{6} + \frac{64}{3} = \frac{64}{6}$$

5. **Ans (4)**

$$y_1 = \frac{dy}{dx} = 2 \left( \ln(x + \sqrt{x^2 + 1}) \right) \cdot \left( \frac{1}{x + \sqrt{x^2 + 1}} \right) \left( 1 + \frac{2x}{2\sqrt{x^2 + 1}} \right)$$

$$= \frac{2 \ln(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}}$$

$$y_1 \sqrt{x^2 + 1} = 2 \ln(x + \sqrt{x^2 + 1})$$

$$y_1^2 (x^2 + 1) = 4 \left( \ln(x + \sqrt{x^2 + 1}) \right)^2$$

$$y_1^2 (x^2 + 1) = 4y$$

diff. w.r.t. x

$$y_1^2 (2x) + (x^2 + 1) \cdot 2y_1 y_2 = 4y_1$$

$$xy_1 + (x^2 + 1)y_2 = 2$$

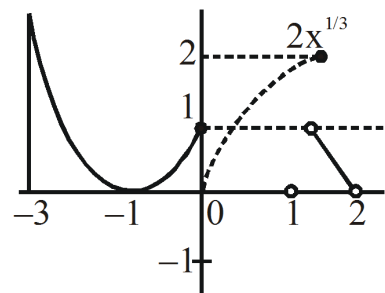
$$xy_2 + y_1 + (x^2 + 1)y_3 + y_2(2x) = 0$$

$$(x^2 + 1) \frac{d^3 y}{dx^3} + (3x) \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0$$

$$p = 3, q = 0, r = 1, s = 0$$

$$\Rightarrow p + q + r + s = 4$$

6. **Ans (5)**



from graph  $-1 \leq a < 1$

$\Rightarrow$  number of integral values of 'a' is 2

(i.e.  $a = -1, 0$ )

and total number of extremas of  $f(x)$  is 3

(i.e.  $x = -1, 0, 1$ )

7. **Ans (189)**

$$\text{Let } F(x) = f^3 + g^3 + h^3 + 3f \cdot g \cdot h$$

$$\Rightarrow F'(x) = 3f^2 f' + 3g^2 g' + 3h^2 h' + 3f'gh + 3fg'h + 3f'gh$$

by putting the given relations we get  $F'(x) = 0$

$$\Rightarrow F(x) = 0 \Rightarrow F(x) = k$$

$$F(0) = 64 + 125 = 189$$

8. **Ans ( 17 )**

$$\frac{dy}{dx} \propto x^3 \Rightarrow \frac{dy}{dx} = kx^3$$

$$y = \frac{kx^4}{4} + c$$

passing through (0,1) &  $\left(\frac{1}{2}, \frac{17}{16}\right)$

$$c = 1, k = 4 \Rightarrow f(x) = x^4 + 1$$

$$\text{length of normal } (\ell) = \frac{17}{16} \sqrt{1 + \left(\frac{4}{8}\right)^2}$$

$$\Rightarrow \frac{17}{16} \frac{\sqrt{5}}{2} \Rightarrow \frac{17\sqrt{5}}{32} \Rightarrow \frac{32\ell}{\sqrt{5}} = 17$$

9. **Ans ( 33 )**

$$f(x) = \frac{100}{3(x^6 + 12x^4 + 5x^2 + 1)}$$

$$1 \leq x^6 + 12x^4 + 5x^2 + 1 < \infty$$

$$\therefore f(x) \in \left(0, \frac{100}{3}\right]$$

$\therefore$  Number of integers = 33

10. **Ans ( 2 )**

$$\int \frac{e^{7x/2}}{\sqrt[4]{e^{2x} + e^{-2x}}} dx$$

$$\int \frac{e^{4x}}{\sqrt[4]{e^{4x} + 1}}, \quad \text{let } e^{4x} + 1 = t \Rightarrow e^{4x} \cdot 4dx = dt$$

$$\frac{1}{4} \int \frac{dt}{t^{1/4}} = \frac{1}{4} \frac{t^{3/4}}{(3/4)} + C$$

$$= \frac{1}{3} (e^{4x} + 1)^{3/4} + C$$

$$\Rightarrow k = 3, \alpha = 4, \beta = 3, \gamma = 4$$

$$\Rightarrow \left(\frac{k}{\beta} + \frac{\gamma}{\alpha}\right) = 2$$